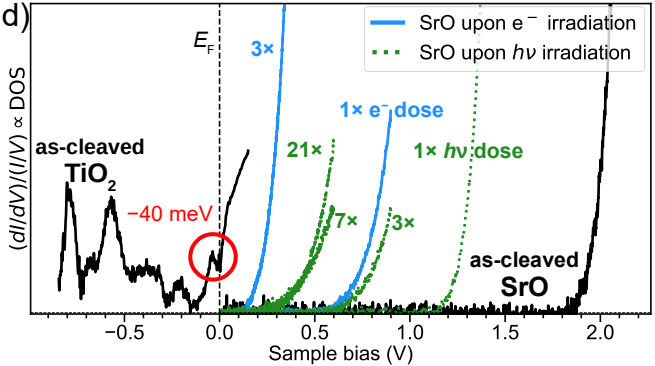
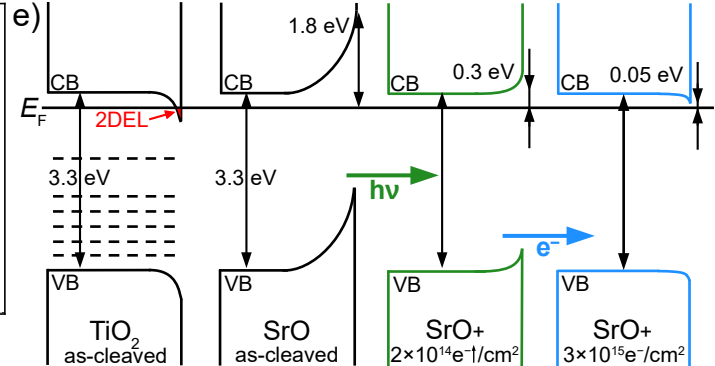

SrO +10^13 e^−/cm^2
a)
SrO +7×10^13 e^−/cm^2
SrO +2×10^14 e^−/cm^2
SrO +10^15 e^− /cm^2
b)
SrO +3×10^15 e^− /cm^2
c)
TiO2 +3×10^15 e^− /cm^2
2 nm
−9 Hz
−4 Hz
#3960
#4016
#4035
#4093
#4160
#4166
−7 Hz
−11 Hz
−10 Hz
−8 Hz
ncAFM (const. z), A=200 pm
ncAFM (const. z), A=250 pm
ncAFM (const. z), A=300 pm


d)
SrO upon e^− irradiation
SrO upon hν irradiation
E_F
3×
1× e^− dose
21×
1× hν dose
7×
3×
as-cleaved TiO2
−40 meV
as-cleaved SrO
(dI/dV)/(I/V) ∝ DOS
Sample bias (V)
−0.5
0.0
0.5
1.0
1.5
2.0


e)
CB
E_F
2DEL
1.8 eV
0.3 eV
0.05 eV
3.3 eV
3.3 eV
hν
e^−
VB
TiO2 as-cleaved
SrO as-cleaved
SrO+ 2×10^14e^−/cm^2
SrO+ 3×10^15e^−/cm^2